Geometric Algebra and Frame Theory

Ian Jorquera
Joint with: Emily J. King

Get the Slides:

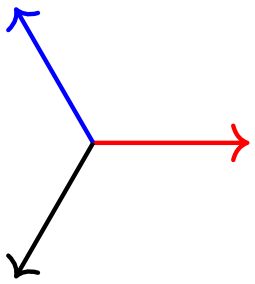
Colorado State University

March 4th, 2026

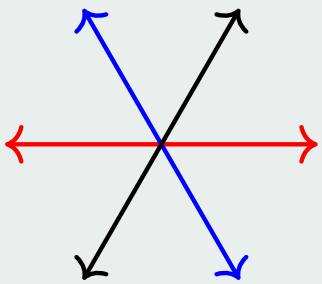
Table of Contents

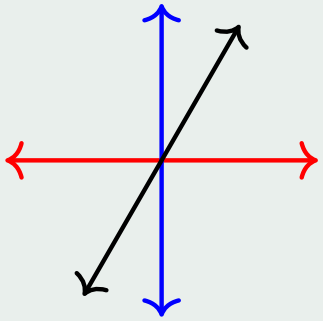
1. Optimal Line Packings Over $\mathbb{R}$ and $\mathbb{C}$
   Welch-Rankin Bound, and Equiangular Tight Frames (ETFs)
2. "Optimal" Line Packings over Finite Fields
3. Equivalence of Line Packings
4. The Welch Bound Revisited

Optimal Line Packings Over $\mathbb{R}$ and $\mathbb{C}$



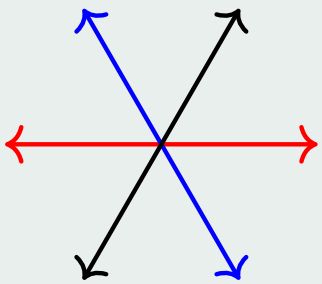
Line Packings: Pack n lines in $\mathbb{R}^d$ or $\mathbb{C}^d$, where every line is maximally spread apart.

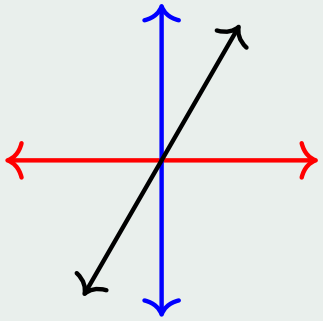




Goal: Maximize pairwise interior angles, or minimize $\cos^2\theta$

Line Packings: Pack n lines in $\mathbb{R}^d$ or $\mathbb{C}^d$, where every line is maximally spread apart.





Goal: Maximize pairwise interior angles, or minimize $\cos^2\theta$

Given n lines, represent each by a unit vector

$$\Phi = \begin{bmatrix} | & | & & | \\ \varphi_1 & \varphi_2 & \cdots & \varphi_n \\ | & | & & | \end{bmatrix} \in \mathbb{F}^{d\times n}$$

New Goal: Minimize $\max_{i\neq j} |\langle \varphi_i, \varphi_j\rangle|^2$

Given n lines in $\mathbb{R}^d$ or $\mathbb{C}^d$, represented each by a unit vector

$$\Phi = \begin{bmatrix} | & | & & | \\ \varphi_1 & \varphi_2 & \cdots & \varphi_n \\ | & | & & | \end{bmatrix}$$

New Goal: Minimize $\max_{i \neq j} |\langle \varphi_i, \varphi_j \rangle|^2$.

Given n lines in $\mathbb{R}^d$ or $\mathbb{C}^d$, represented each by a unit vector

$$\Phi = \begin{bmatrix} | & | & & | \\ \varphi_1 & \varphi_2 & \cdots & \varphi_n \\ | & | & & | \end{bmatrix}$$

New Goal: Minimize $\max_{i \neq j} |\langle \varphi_i, \varphi_j \rangle|^2$.

Coherence: $\mu^2(\Phi) = \max_{i \neq j} |\langle \varphi_i, \varphi_j \rangle|^2 \geq 0$

Φ is a **Grassmannian Frame** if Φ is a global minimizer for the coherence

Given n lines in $\mathbb{R}^d$ or $\mathbb{C}^d$, represented each by a unit vector

$$\Phi = \begin{bmatrix} | & | & & | \\ \varphi_1 & \varphi_2 & \cdots & \varphi_n \\ | & | & & | \end{bmatrix}$$

New Goal: Minimize $\max_{i \neq j} |\langle \varphi_i, \varphi_j \rangle|^2$.

Coherence: $\mu^2(\Phi) = \max_{i \neq j} |\langle \varphi_i, \varphi_j \rangle|^2 \geq 0$

Φ is a **Grassmannian Frame** if Φ is a global minimizer for the coherence

How To Find Grassmannian Frames:

Step 1: Find a lower bound on coherence.

Step 2: Find examples which meet the bound.

Given n lines in $\mathbb{R}^d$ or $\mathbb{C}^d$, represented each by a unit vector

$$\Phi = \begin{bmatrix} | & | & & | \\ \varphi_1 & \varphi_2 & \cdots & \varphi_n \\ | & | & & | \end{bmatrix}$$

New Goal: Minimize $\max_{i \neq j} |\langle \varphi_i, \varphi_j \rangle|^2$.

Coherence: $\mu^2(\Phi) = \max_{i \neq j} |\langle \varphi_i, \varphi_j \rangle|^2 \geq 0$

Φ is a **Grassmannian Frame** if Φ is a global minimizer for the coherence

How To Find Grassmannian Frames:

Step 1: Find a lower bound on coherence.
Step 2: Find examples which meet the bound.

$\mu^2(\Phi) \geq 0 \Rightarrow \Phi = (\varphi_j)_{j=1}^n$ orthonormal is a Grassmannian frame.

The Welch-Rankin Bound and Equinagular Tight Frames.

For $\Phi = \left[\varphi_1, \ldots, \varphi_n\right]$ in $\mathbb{F}^d$.

Welch-Rankin Bound (Welch; 1974) (Rankin; 1955)

$$\mu^2(\Phi) = \max_{i \neq j} |\langle \varphi_i, \varphi_j \rangle|^2 \geq \frac{n-d}{d(n-1)}$$

The Welch-Rankin Bound and Equinagular Tight Frames.

For $\Phi = \left[\varphi_1, \ldots, \varphi_n\right]$ in $\mathbb{F}^d$.

Welch-Rankin Bound (Welch; 1974) (Rankin; 1955)

$$\mu^2(\Phi) = \max_{i \neq j} |\langle \varphi_i, \varphi_j \rangle|^2 \geq \frac{n-d}{d(n-1)}$$

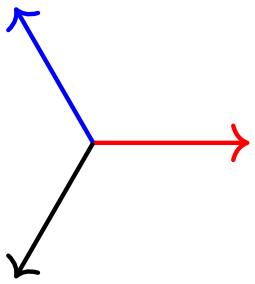
With equality if and only if

$$\left.\begin{array}{l} \bullet\ \textbf{Equiangular}\text{: } |\langle \varphi_i, \varphi_j \rangle|^2 = b \text{ for all } i \neq j \\ \bullet\ \textbf{Tightness}\text{: } \Phi\Phi^* = cI \end{array}\right\} \Phi \text{ is an ETF}$$

Tightness generalize the Pythagorean theorem or Parceval's identity

$$\Phi\Phi^* = cI \Leftrightarrow \sum_{i=1}^{d} \left\|\langle x, \varphi_i \rangle \varphi_i\right\|^2 = c\left\|x\right\|^2$$

"Optimal" Line Packings over Finite Fields



Discretizing Reality: Finite Field Analog to $\mathbb{R}^d$

Real IP Spaces	$\rightsquigarrow$	**Orthogonal Geometries**
$\mathbb{R}^d$	$\rightsquigarrow$	$\mathbb{F}_q^d$, where $q = p^\ell$ is odd.
Inner Products	$\rightsquigarrow$	**Non-Degenerate Symmetric Scalar Products**

Discretizing Reality: Finite Field Analog to $\mathbb{R}^d$

Real IP Spaces	$\rightsquigarrow$	**Orthogonal Geometries**
$\mathbb{R}^d$	$\rightsquigarrow$	$\mathbb{F}_q^d$, where $q = p^\ell$ is odd.
Inner Products	$\rightsquigarrow$	**Non-Degenerate Symmetric Scalar Products**
$\langle -, - \rangle : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ $\langle x, y \rangle = \langle y, x \rangle$ $\langle x, - \rangle : \mathbb{R}^d \to \mathbb{R}$ linear		

Discretizing Reality: Finite Field Analog to $\mathbb{R}^d$

Real IP Spaces	$\rightsquigarrow$	**Orthogonal Geometries**
$\mathbb{R}^d$	$\rightsquigarrow$	$\mathbb{F}_q^d$, where $q = p^\ell$ is odd.
Inner Products	$\rightsquigarrow$	**Non-Degenerate Symmetric Scalar Products**
$\langle -, - \rangle : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ $\langle x, y \rangle = \langle y, x \rangle$ $\langle x, - \rangle : \mathbb{R}^d \to \mathbb{R}$ linear $\langle x, x \rangle > 0$ iff $x \neq 0$		

Discretizing Reality: Finite Field Analog to $\mathbb{R}^d$

Real IP Spaces	$\rightsquigarrow$	**Orthogonal Geometries**
$\mathbb{R}^d$	$\rightsquigarrow$	$\mathbb{F}_q^d$, where $q = p^\ell$ is odd.
Inner Products	$\rightsquigarrow$	**Non-Degenerate Symmetric Scalar Products**
$\langle -, - \rangle : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$		$\langle -, - \rangle : \mathbb{F}_q^d \times \mathbb{F}_q^d \to \mathbb{F}_q$
$\langle x, y \rangle = \langle y, x \rangle$	$\rightsquigarrow$	$\langle x, y \rangle = \langle y, x \rangle$
$\langle x, - \rangle : \mathbb{R}^d \to \mathbb{R}$ linear		$\langle x, - \rangle : \mathbb{F}_q^d \to \mathbb{F}_q$ linear
$\langle x, x \rangle > 0$ iff $x \neq 0$		

Discretizing Reality: Finite Field Analog to $\mathbb{R}^d$

Real IP Spaces	$\rightsquigarrow$	**Orthogonal Geometries**
$\mathbb{R}^d$	$\rightsquigarrow$	$\mathbb{F}_q^d$, where $q = p^\ell$ is odd.
Inner Products	$\rightsquigarrow$	**Non-Degenerate Symmetric Scalar Products**
$\langle -, - \rangle : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$		$\langle -, - \rangle : \mathbb{F}_q^d \times \mathbb{F}_q^d \to \mathbb{F}_q$
$\langle x, y \rangle = \langle y, x \rangle$	$\rightsquigarrow$	$\langle x, y \rangle = \langle y, x \rangle$
$\langle x, - \rangle : \mathbb{R}^d \to \mathbb{R}$ linear		$\langle x, - \rangle : \mathbb{F}_q^d \to \mathbb{F}_q$ linear
$\langle x, x \rangle > 0$ iff $x \neq 0$	$\rightsquigarrow$	$\langle x, y \rangle \neq 0$ for some $y \in \mathbb{F}_q^d$ iff $x \neq 0$

Discretizing Reality: Finite Field Analog to $\mathbb{R}^d$

Real IP Spaces	$\rightsquigarrow$	**Orthogonal Geometries**
$\mathbb{R}^d$	$\rightsquigarrow$	$\mathbb{F}_q^d$, where $q = p^\ell$ is odd.
Inner Products	$\rightsquigarrow$	**Non-Degenerate Symmetric Scalar Products**
$\langle -, - \rangle : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$		$\langle -, - \rangle : \mathbb{F}_q^d \times \mathbb{F}_q^d \to \mathbb{F}_q$
$\langle x, y \rangle = \langle y, x \rangle$	$\rightsquigarrow$	$\langle x, y \rangle = \langle y, x \rangle$
$\langle x, - \rangle : \mathbb{R}^d \to \mathbb{R}$ linear		$\langle x, - \rangle : \mathbb{F}_q^d \to \mathbb{F}_q$ linear
$\langle x, x \rangle > 0$ iff $x \neq 0$	$\rightsquigarrow$	$\langle x, y \rangle \neq 0$ for some $y \in \mathbb{F}_q^d$ iff $x \neq 0$

Example: Non-degeneracy as a proof of being non-zero

$V = \mathbb{F}_3^3$ with $\langle x, y \rangle = x^\intercal y$ the dot product.

$$x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \qquad \left\langle \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\rangle = 0 \qquad \left\langle \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\rangle = 1$$

Discretizing the Imaginary: Finite Field Analog to $\mathbb{C}^d$

Complex IP Spaces	$\rightsquigarrow$	**Unitary Geometries**
$\mathbb{C}^d$	$\rightsquigarrow$	$\mathbb{F}_{q^2}^d$, where $q = p^\ell$.
$x \mapsto \overline{x}$	$\rightsquigarrow$	$x \mapsto x^q$
Inner Products	$\rightsquigarrow$	**Non-Degenerate Hermitian Scalar Products**

Discretizing the Imaginary: Finite Field Analog to $\mathbb{C}^d$

Complex IP Spaces	$\rightsquigarrow$	**Unitary Geometries**
$\mathbb{C}^d$	$\rightsquigarrow$	$\mathbb{F}_{q^2}^d$, where $q = p^\ell$.
$x \mapsto \overline{x}$	$\rightsquigarrow$	$x \mapsto x^q$

Inner Products	$\rightsquigarrow$	**Non-Degenerate Hermitian Scalar Products**
$\langle -, - \rangle : \mathbb{C}^d \times \mathbb{C}^d \to \mathbb{C}$		$\langle -, - \rangle : \mathbb{F}_{q^2}^d \times \mathbb{F}_{q^2}^d \to \mathbb{F}_{q^2}$
$\boldsymbol{\langle x, y \rangle = \overline{\langle y, x \rangle}}$	$\rightsquigarrow$	$\boldsymbol{\langle x, y \rangle = \langle y, x \rangle^q}$
$\langle x, - \rangle : \mathbb{C}^d \to \mathbb{C}$ linear		$\langle x, - \rangle : \mathbb{F}_{q^2}^d \to \mathbb{F}_{q^2}$ linear
$\langle x, x \rangle > 0$ iff $x \neq 0$	$\rightsquigarrow$	$\langle x, y \rangle \neq 0$ for some $y \in \mathbb{F}_{q^2}^d$ iff $x \neq 0$

Geometric Algebra (not algebraic geometry)

A $\mathbb{F}$-vector space V is called **non-degenerate** if it has a non-degenerate symmetric/Hermitian scalar product.

Geometric Algebra (not algebraic geometry)

A $\mathbb{F}$-vector space V is called **non-degenerate** if it has a non-degenerate symmetric/Hermitian scalar product.

Case O ($\mathbb{F} = \mathbb{F}_q$)

$V = \mathbb{F}_q^d$ (choosing a basis)
$\langle x, y \rangle = x^\intercal M y$, where $M^\intercal = M$ and is invertible.

There is a basis for V such that $M = \mathrm{Diag}(1, \ldots, 1, \delta)$

- $\delta = 1$ is a square
- δ is not a square

Geometric Algebra (not algebraic geometry)

A $\mathbb{F}$-vector space V is called **non-degenerate** if it has a non-degenerate symmetric/Hermitian scalar product.

Case O ($\mathbb{F} = \mathbb{F}_q$)

$V = \mathbb{F}_q^d$ (choosing a basis)
$\langle x, y \rangle = x^\intercal M y$, where $M^\intercal = M$ and is invertible.

There is a basis for V such that $M = \mathrm{Diag}(1, \ldots, 1, \delta)$

- $\delta = 1$ is a square
- δ is not a square

Case U ($\mathbb{F} = \mathbb{F}_{q^2}$)

$V = \mathbb{F}_{q^2}^d$ (choosing a basis)
$\langle x, y \rangle = x^* M y$, where $M^* = M$ and is invertible.

There is a basis for V such that $M = \mathrm{Diag}(1, \ldots, 1)$

Geometric Algebra (not algebraic geometry)

A $\mathbb{F}$-vector space V is called **non-degenerate** if it has a non-degenerate symmetric/Hermitian scalar product.

Case O ($\mathbb{F} = \mathbb{F}_q$)

$V = \mathbb{F}_q^d$ (choosing a basis)
$\langle x, y \rangle = x^\intercal M y$, where $M^\intercal = M$ and is invertible.

There is a basis for V such that $M = \mathrm{Diag}(1, \ldots, 1, \delta)$

- $\delta = 1$ is a square
- δ is not a square

Case U ($\mathbb{F} = \mathbb{F}_{q^2}$)

$V = \mathbb{F}_{q^2}^d$ (choosing a basis)
$\langle x, y \rangle = x^* M y$, where $M^* = M$ and is invertible.

There is a basis for V such that $M = \mathrm{Diag}(1, \ldots, 1)$

The **adjoint** of A, is the map $A^\dagger$ such that $\langle Ax, y \rangle = \langle x, A^\dagger y \rangle$.

A map $A : V \to V$ is called a **unitary** if $\langle Ax, Ay \rangle = \langle x, y \rangle$ ($A^\dagger A = I$)

Types of Geometry over $\mathbb{R}$ and $\mathbb{C}$

The Real Case

$V = \mathbb{R}^d$
$\langle x, y \rangle = x^\intercal M y$, where $M^\intercal = M$ and is invertible.

There is a basis for V such that $M = \text{Diag}(1, \ldots, 1, -1, \ldots, -1)$

If $M = \text{Diag}(1, \ldots, 1)$, then $\langle x, y \rangle$ is an Inner product.

The Complex Case

$V = \mathbb{C}^d$
$\langle x, y \rangle = x^* M y$, where $M^* = M$ and is invertible.

There is a basis for V such that $M = \text{Diag}(1, \ldots, 1)$, and $\langle x, y \rangle$ is an inner product.

For inner products: $A^\dagger = A^*$.

What is an Elephant Really?

What is an Elephant Really?

Inner Product Spaces:

- Subspaces of inner product spaces are inner product spaces
- $\Phi = \left[\varphi_1, \ldots, \varphi_n\right]$ and its Gram matrix $\Phi^*\Phi = [\langle \varphi_i, \varphi_j \rangle]$ give "equivalent information".

What is an Elephant Really?

Inner Product Spaces:

- Subspaces of inner product spaces are inner product spaces
- $\Phi = \left[\varphi_1, \ldots, \varphi_n\right]$ and its Gram matrix $\Phi^*\Phi = [\langle \varphi_i, \varphi_j \rangle]$ give "equivalent information".

Case O and U: Not the case. Consider an orthogonal geometry $V = \mathbb{F}_3^4$ with $\langle x, y \rangle = x^\intercal y$

$$\Phi = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 2 & 2 \\ 0 & 1 \end{bmatrix} \quad \Phi^\dagger \Phi = [\langle \varphi_i, \varphi_j \rangle] = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$\operatorname{im} \Phi \subseteq V$ is degenerate.

Frame Theory (Greaves, Iverson, Jasper, Mixon; 2022), (**J**, King; 2025)

Let $\Phi = \left[\varphi_1, \varphi_2 \ldots, \varphi_n\right]$ from $V = \mathbb{F}^d$, $a, b, c \in \mathbb{F}$. Then Φ is a

- **frame** for $\operatorname{im}\Phi$ if $\operatorname{im}\Phi$ is non-degenerate $\Leftrightarrow$ $\mathrm{rk}(\Phi) = \mathrm{rk}(\Phi^\dagger\Phi)$
- c-**tight frame** for $\operatorname{im}\Phi$ if $\Phi(\Phi^\dagger\Phi) = c\Phi$
- (a, b)-**equiangular** if
  - $\langle\varphi_j, \varphi_j\rangle = a$ for all j
  - $\langle\varphi_j, \varphi_k\rangle\langle\varphi_k, \varphi_j\rangle = b$ for all $j \neq k$
- (a, b, c)-**equiangular tight frame(ETF)** if all the above.

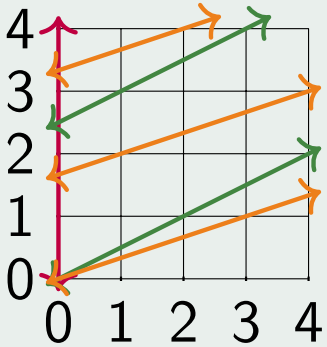
Frame Theory (Greaves, Iverson, Jasper, Mixon; 2022), (**J**, King; 2025)

Let $\Phi = \left[\varphi_1, \varphi_2 \ldots, \varphi_n\right]$ from $V = \mathbb{F}^d$, $a, b, c \in \mathbb{F}$. Then Φ is a

- **frame** for $\operatorname{im}\Phi$ if $\operatorname{im}\Phi$ is non-degenerate $\Leftrightarrow$ $\operatorname{rk}(\Phi) = \operatorname{rk}(\Phi^\dagger\Phi)$
- c-**tight frame** for $\operatorname{im}\Phi$ if $\Phi(\Phi^\dagger\Phi) = c\Phi$
- (a, b)-**equiangular** if
  - $\langle\varphi_j, \varphi_j\rangle = a$ for all j
  - $\langle\varphi_j, \varphi_k\rangle\langle\varphi_k, \varphi_j\rangle = b$ for all $j \neq k$
- (a, b, c)-**equiangular tight frame(ETF)** if all the above.

Example: $V = \mathbb{F}_5^2$ with $\langle x, y\rangle = x^\intercal My$, where $M = \operatorname{Diag}(1, 3)$

$$\Phi = \begin{bmatrix} 0 & 2 & 3 \\ 2 & 1 & 1 \end{bmatrix} \qquad \Phi^\dagger\Phi = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$



Φ is an $(2, 1, 3)$-ETF for $\mathbb{F}_5^2$ of $n = 3$ vectors.

Frame Theory: ETFs in case O and U

$V = \mathbb{F}_3^4$ with $\langle x, y \rangle = x^\intercal M y$, where $M = \mathrm{Diag}(1, 1, 1, 2)$

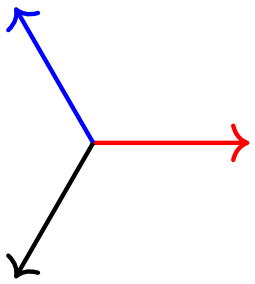
$$\Phi = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 2 & 1 & 2 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Φ is an $(0, 1, 0)$-ETF for $\mathbb{F}_3^4$ of $n = 10$ vectors.

$V = \mathbb{F}_{3^2}^5$ with $\langle x, y \rangle = x^* y$. Ψ is a $(0, 1, 0)$-ETF of 16 vectors.

$$\Psi = \begin{bmatrix} 1 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & a & a & a & a & a^3 & a^3 & a^3 & a^3 \\ a & a & a^5 & a^5 & a^5 & a^5 & a^5 & a^5 & 1 & 1 & 1 & 1 & a^6 & a^6 & a^6 & a^6 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2 & a & a & a^5 & a^5 & a^3 & a^3 & a^7 & a^7 \\ 0 & 0 & 0 & 0 & 1 & 2 & 0 & 0 & a & a^5 & a & a^5 & a^3 & a^7 & a^3 & a^7 \\ 0 & 0 & a^2 & a^6 & 0 & 0 & 0 & 0 & a^7 & a^3 & a^3 & a^7 & a & a^5 & a^5 & a \end{bmatrix}$$

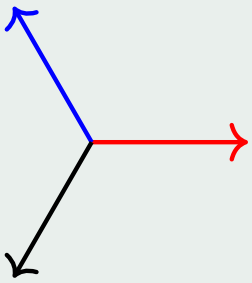
Equivalence of Line Packings

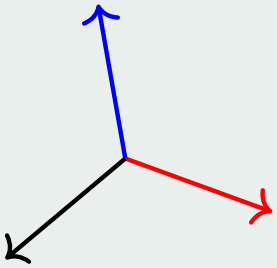


When are systems of lines the "same"? (**J**, King; 2025)

Let $\Phi = (\varphi_j)_{j=1}^n$ and $\Psi = (\psi_j)_{j=1}^n$ be collections of lines in $\mathbb{F}^d$.

Unitary Equivalence: if $\Psi = U\Phi$, $U \in U(\mathbb{F}^d)$. $\Rightarrow$ $\Psi^\dagger\Psi = \Phi^\dagger\Phi$

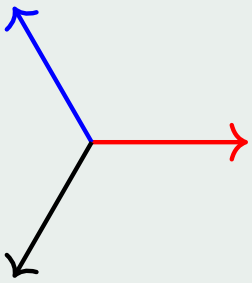
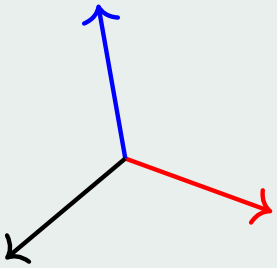
$\Phi =$ 

$\Psi =$ 

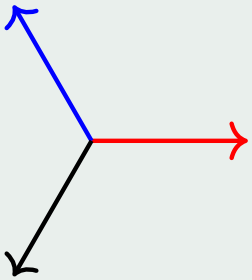
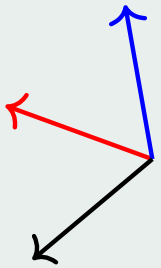
When are systems of lines the "same"? (**J**, King; 2025)

Let $\Phi = (\varphi_j)_{j=1}^n$ and $\Psi = (\psi_j)_{j=1}^n$ be collections of lines in $\mathbb{F}^d$.

Unitary Equivalence: if $\Psi = U\Phi$, $U \in U(\mathbb{F}^d)$. $\Rightarrow$ $\Psi^\dagger\Psi = \Phi^\dagger\Phi$

$\Phi =$  $\Psi =$ 

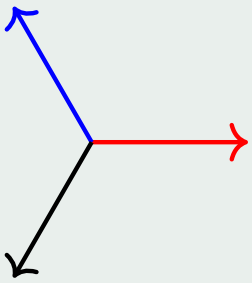
Switching Equivalence: if $\Psi = U\Phi T$ where $T = \mathrm{diag}(t_1, \ldots, t_n)$, and $|d_i|^2 = 1$ $\Rightarrow$ $\Psi^\dagger\Psi = T^\dagger\Phi^\dagger\Phi T$

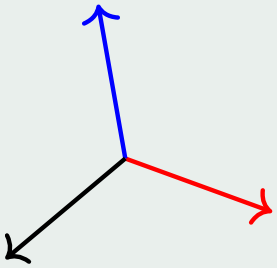
$\Phi =$  $\Psi =$ 

When are systems of lines the "same"? (**J**, King; 2025)

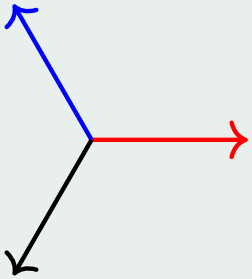
Let $\Phi = (\varphi_j)_{j=1}^n$ and $\Psi = (\psi_j)_{j=1}^n$ be frames for $\mathbb{F}^d$.

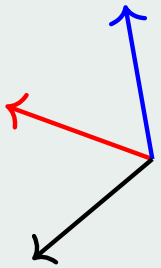
Unitary Equivalence: if $\Psi = U\Phi$, $U \in U(\mathbb{F}^d)$. $\Leftrightarrow$ $\Psi^\dagger\Psi = \Phi^\dagger\Phi$

$\Phi =$ 

$\Psi =$ 

Switching Equivalence: if $\Psi = U\Phi T$ where $T = \mathrm{diag}(t_1, \ldots, t_n)$, and $|d_i|^2 = 1$ $\Leftrightarrow$ $\Psi^\dagger\Psi = T^\dagger\Phi^\dagger\Phi T$

$\Phi =$ 

$\Psi =$ 

Invariants of Equivalence

Let $\Phi = (\varphi_j)_{j=1}^n$ and $\Psi = (\psi_j)_{j=1}^n$ be frames for $\mathbb{F}^d$.

m-product (m-vertex Bargmann invariant)

$\Delta(\varphi_{j_1}, \varphi_{j_2}, \ldots, \varphi_{j_m}) = \langle \varphi_{j_1}, \varphi_{j_2} \rangle \langle \varphi_{j_2}, \varphi_{j_3} \rangle \cdots \langle \varphi_{j_{m-1}}, \varphi_{j_m} \rangle \langle \varphi_{j_m}, \varphi_{j_1} \rangle$

Double Product: $\Delta(\varphi_j, \varphi_k) = \langle \varphi_j, \varphi_k \rangle \langle \varphi_k, \varphi_j \rangle$

Triple Product: $\Delta(\varphi_j, \varphi_k, \varphi_\ell) = \langle \varphi_j, \varphi_k \rangle \langle \varphi_k, \varphi_\ell \rangle \langle \varphi_\ell, \varphi_j \rangle$

Invariants of Equivalence

Let $\Phi = (\varphi_j)_{j=1}^n$ and $\Psi = (\psi_j)_{j=1}^n$ be frames for $\mathbb{F}^d$.

m-product (m-vertex Bargmann invariant)

$\Delta(\varphi_{j_1}, \varphi_{j_2}, \ldots, \varphi_{j_m}) = \langle \varphi_{j_1}, \varphi_{j_2} \rangle \langle \varphi_{j_2}, \varphi_{j_3} \rangle \cdots \langle \varphi_{j_{m-1}}, \varphi_{j_m} \rangle \langle \varphi_{j_m}, \varphi_{j_1} \rangle$

Double Product: $\Delta(\varphi_j, \varphi_k) = \langle \varphi_j, \varphi_k \rangle \langle \varphi_k, \varphi_j \rangle$

Triple Product: $\Delta(\varphi_j, \varphi_k, \varphi_\ell) = \langle \varphi_j, \varphi_k \rangle \langle \varphi_k, \varphi_\ell \rangle \langle \varphi_\ell, \varphi_j \rangle$

If Φ and Ψ switching equivalent ($\Psi = U\Phi T$) then

- $\Delta(\varphi_j) = \Delta(\psi_j)$
- $\Delta(\varphi_j, \varphi_k) = \Delta(\psi_j, \psi_k)$
- $\Delta(\varphi_j, \varphi_k, \varphi_\ell) = \Delta(\psi_j, \psi_k, \psi_\ell)$
- $\ldots$
- $\Delta(\varphi_{j_1}, \varphi_{j_2}, \ldots, \varphi_{j_n}) = \Delta(\psi_{j_1}, \psi_{j_2}, \ldots, \psi_{j_n})$

Invariants of Equivalence

Let $\Phi = (\varphi_j)_{j=1}^n$ and $\Psi = (\psi_j)_{j=1}^n$ be frames for $\mathbb{F}^d$.

m-product (m-vertex Bargmann invariant)

$\Delta(\varphi_{j_1}, \varphi_{j_2}, \ldots, \varphi_{j_m}) = \langle \varphi_{j_1}, \varphi_{j_2} \rangle \langle \varphi_{j_2}, \varphi_{j_3} \rangle \cdots \langle \varphi_{j_{m-1}}, \varphi_{j_m} \rangle \langle \varphi_{j_m}, \varphi_{j_1} \rangle$

Double Product: $\Delta(\varphi_j, \varphi_k) = \langle \varphi_j, \varphi_k \rangle \langle \varphi_k, \varphi_j \rangle$

Triple Product: $\Delta(\varphi_j, \varphi_k, \varphi_\ell) = \langle \varphi_j, \varphi_k \rangle \langle \varphi_k, \varphi_\ell \rangle \langle \varphi_\ell, \varphi_j \rangle$

If Φ and Ψ switching equivalent ($\Psi = U\Phi T$) then

- $\Delta(\varphi_j) = \Delta(\psi_j)$
- $\Delta(\varphi_j, \varphi_k) = \Delta(\psi_j, \psi_k)$
- $\Delta(\varphi_j, \varphi_k, \varphi_\ell) = \Delta(\psi_j, \psi_k, \psi_\ell)$
- $\ldots$
- $\Delta(\varphi_{j_1}, \varphi_{j_2}, \ldots, \varphi_{j_n}) = \Delta(\psi_{j_1}, \psi_{j_2}, \ldots, \psi_{j_n})$

Theorem (Gallagher, Proulx; 1977), (Chien, Waldron; 2016), (J)

If $\Delta(\varphi_{j_1}, \varphi_{j_2}, \ldots, \varphi_{j_m}) = \Delta(\psi_{j_1}, \psi_{j_2}, \ldots, \psi_{j_m})$ for all $1 \leq m \leq n$ then Φ and Ψ switching equivalent

We can do better! 3 is plenty!

Let $\Phi = (\varphi_j)_{j=1}^n$ and $\Psi = (\psi_j)_{j=1}^n$ be frame for $\mathbb{F}^d$.

Theorem (J, King; 2025)

If for all $j \neq k$ we have that $\langle \varphi_j, \varphi_k \rangle \neq 0$ and $\langle \psi_j, \psi_k \rangle \neq 0$ then Φ and Ψ are switching equivalent if and only if all double and triple products agree.

We can do better! 3 is plenty!

Let $\Phi = (\varphi_j)_{j=1}^n$ and $\Psi = (\psi_j)_{j=1}^n$ be frame for $\mathbb{F}^d$.

Theorem (J, King; 2025)

If for all $j \neq k$ we have that $\langle \varphi_j, \varphi_k \rangle \neq 0$ and $\langle \psi_j, \psi_k \rangle \neq 0$ then Φ and Ψ are switching equivalent if and only if all double and triple products agree.

Theorem (J, King; 2025)

If Φ and Ψ are (a, b)-equiangular frames (magnitude of every vector is a and all double products equal b) then Φ and Ψ are switching equivalent if and only if all triple products agree.

We can do better! 3 is plenty!

Let $\Phi = (\varphi_j)_{j=1}^n$ and $\Psi = (\psi_j)_{j=1}^n$ be frame for $\mathbb{F}^d$.

Theorem (J, King; 2025)

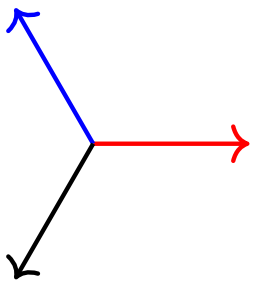
If for all $j \neq k$ we have that $\langle \varphi_j, \varphi_k \rangle \neq 0$ and $\langle \psi_j, \psi_k \rangle \neq 0$ then Φ and Ψ are switching equivalent if and only if all double and triple products agree.

Theorem (J, King; 2025)

If Φ and Ψ are (a, b)-equiangular frames (magnitude of every vector is a and all double products equal b) then Φ and Ψ are switching equivalent if and only if all triple products agree.

For this reason triple products have been widely used to study the algebraic properties of frames. See (Appleby et. al.; 2011), (Zhu; 2015), and (King; 2019).

The Welch Bound Revisited



A Flash Back: Understanding ETFs

For $\Phi = [\varphi_1, \ldots, \varphi_n]$ in $\mathbb{F}^d$.

Welch-Rankin Bound (Welch; 1974) (Rankin; 1955)

$$\mu^2(\Phi) = \max_{i \neq j} |\langle \varphi_i, \varphi_j \rangle|^2 \geq \frac{n-d}{d(n-1)}$$

With equality if and only if

$$\left.\begin{array}{l} \bullet\ \textbf{Equiangular}\text{: } |\langle \varphi_i, \varphi_j \rangle|^2 = b \text{ for all } i \neq j \\ \bullet\ \textbf{Tightness}\text{: } \Phi\Phi^* = cI \end{array}\right\} \Phi \text{ is an ETF}$$

A Flash Back: Understanding ETFs

ETFs (optimal line packings) are understood in two ways

Geometrically as ETFs

- **Equiangular**: $i \neq j$
  $|\langle \varphi_i, \varphi_j \rangle|^2 = b$
- **Tightness**: $\Phi\Phi^* = cI$

Combinatorially with

$$b = \frac{n-d}{d(n-1)}$$

- $n = \#$ lines
- $d =$ dimension

Do we get this over Finite Fields?
Short answer: No.
Long answer: Sort of!

On the Failure of a Welch-Rankin Equality

Theorem (Greaves, Iverson, Jasper, Mixon; 2022)

If Φ is a (a, b, c)-ETF for $V = \mathbb{F}^d$ then $d(n-1)b \equiv (n-d)a^2$
(if the field is nice: $b = \frac{n-d}{d(n-1)}a^2$)

On the Failure of a Welch-Rankin Equality

Theorem (Greaves, Iverson, Jasper, Mixon; 2022)

If Φ is a (a, b, c)-ETF for $V = \mathbb{F}^d$ then $d(n-1)b \equiv (n-d)a^2$
(if the field is nice: $b = \frac{n-d}{d(n-1)}a^2$)

Example: $V = \mathbb{F}_5^7$ with $\langle x, y \rangle = x^\intercal y$

$$\Phi = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 & 4 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 & 4 & 0 & 2 \\ 0 & 1 & 1 & 2 & 1 & 2 & 2 & 3 \\ 1 & 0 & 1 & 2 & 3 & 2 & 2 & 3 \\ 1 & 1 & 0 & 2 & 3 & 4 & 4 & 1 \end{bmatrix}$$

Φ is an $(2, 1)$-equiangular frame for V.
It satisfies $b \equiv 1 \equiv \frac{1}{49}2^2 \equiv \frac{n-d}{d(n-1)}a^2$.

On the Failure of a Welch-Rankin Equality

Theorem (Greaves, Iverson, Jasper, Mixon; 2022)

If Φ is a (a, b, c)-ETF for $V = \mathbb{F}^d$ then $d(n-1)b \equiv (n-d)a^2$
(if the field is nice: $b = \frac{n-d}{d(n-1)}a^2$)

Example: $V = \mathbb{F}_5^7$ with $\langle x, y \rangle = x^\intercal y$

$$\Phi = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 & 4 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 & 4 & 0 & 2 \\ 0 & 1 & 1 & 2 & 1 & 2 & 2 & 3 \\ 1 & 0 & 1 & 2 & 3 & 2 & 2 & 3 \\ 1 & 1 & 0 & 2 & 3 & 4 & 4 & 1 \end{bmatrix}$$

Φ is an $(2, 1)$-equiangular frame for V.
It satisfies $b \equiv 1 \equiv \frac{1}{49}2^2 \equiv \frac{n-d}{d(n-1)}a^2$. But Φ is not a tight frame

A New Hope: Using Sums of Triple Products

Let $\mathbb{F}$ be a field with $\mathrm{char}\mathbb{F} \nmid dn$, and $V = \mathbb{F}^d$ in case O or U.

Theorem (J; 2025)

Let $\Phi = \left[\varphi_1, \ldots, \varphi_n\right]$ for V be an (a, b)-equiangular frame for V $(a \neq 0)$. Then Φ is an $(a, b, na/d)$-ETF if and only if

-
-

A New Hope: Using Sums of Triple Products

Let $\mathbb{F}$ be a field with $\mathrm{char}\mathbb{F} \nmid dn$, and $V = \mathbb{F}^d$ in case O or U.

Theorem (J; 2025)

Let $\Phi = \left[\varphi_1, \ldots, \varphi_n\right]$ for V be an (a, b)-equiangular frame for V $(a \neq 0)$. Then Φ is an $(a, b, na/d)$-ETF if and only if

- $d(n-1)b = (n-d)a^2$
-

A New Hope: Using Sums of Triple Products

Let $\mathbb{F}$ be a field with $\mathrm{char}\mathbb{F} \nmid dn$, and $V = \mathbb{F}^d$ in case O or U.

Theorem (J; 2025)

Let $\Phi = \left[\varphi_1, \ldots, \varphi_n\right]$ for V be an (a, b)-equiangular frame for V $(a \neq 0)$. Then Φ is an $(a, b, na/d)$-ETF if and only if

- $d(n-1)b = (n-d)a^2$
- $\displaystyle\sum_{\ell=1}^{n} \langle \varphi_j, \varphi_k \rangle \langle \varphi_k, \varphi_\ell \rangle \langle \varphi_\ell, \varphi_j \rangle = \frac{nab}{d}$ for all $j \neq k$

Applications of this Welch-Rankin Equality

Theorem (J; 2025)

Let $\Phi = \left[\varphi_1, \ldots, \varphi_n\right]$ for $\mathbb{F}^d$ be an (a, b)-equiangular frame

- $\mathrm{char}\mathbb{F} \nmid d(d+1)$, and $a \neq 0$, $a^2 \neq b$
- $\Gamma \subseteq \Phi$ of $d+1$ vectors, where $\langle \varphi_j, \varphi_k \rangle \langle \varphi_k, \varphi_\ell \rangle \langle \varphi_\ell, \varphi_j \rangle$ is constant for all distinct $\varphi_j, \varphi_k, \varphi_\ell \in \Gamma$

Then Γ is a regular d-simplex, an $(a, b, \frac{(d+1)a}{d})$-ETF

The converse of this is also true: a d-simplex has equal triple products.

Applications of this Welch-Rankin Equality

Theorem (J; 2025)

Let $\Phi = \left[\varphi_1, \ldots, \varphi_n\right]$ for $\mathbb{F}^d$ be an (a, b)-equiangular frame

- $\mathrm{char}\mathbb{F} \nmid d(d+1)$, and $a \neq 0$, $a^2 \neq b$
- $\Gamma \subseteq \Phi$ of $d+1$ vectors, where $\langle \varphi_j, \varphi_k \rangle \langle \varphi_k, \varphi_\ell \rangle \langle \varphi_\ell, \varphi_j \rangle$ is constant for all distinct $\varphi_j, \varphi_k, \varphi_\ell \in \Gamma$

Then Γ is a regular d-simplex, an $(a, b, \frac{(d+1)a}{d})$-ETF

The converse of this is also true: a d-simplex has equal triple products.

More Results (J; 2025)

- Determine existence of k-simplices contained in Φ (which span subspaces when $k < d$).
- Show that certain, case O, ETFs with $n = d(d+1)/2$ vectors give rise to combinatorial 4-designs.